Project: Worked examples
Title: Lifting lug and pin joint
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the net-section tension, bearing, tear-out, pin shear and pin bending of a pinned lifting lug between two cheek plates under a load along the lug. It is a design calculation, not a thorough examination.

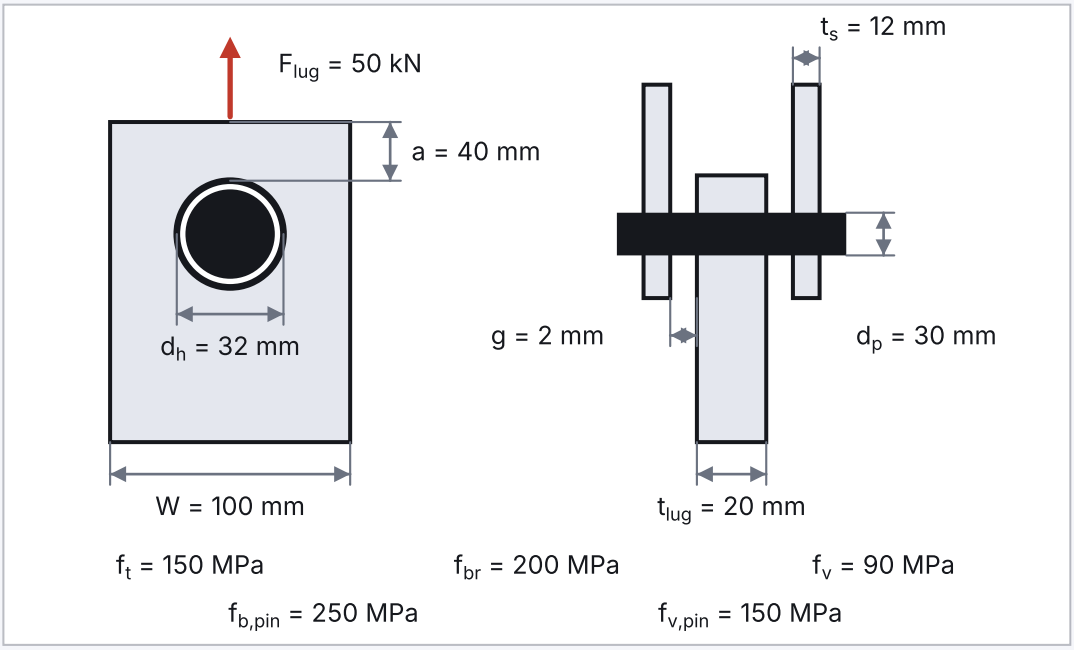


Figure 1 — A lug and its pin, front view and section through the pin: the lug between two cheek plates, the load through the pin. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Net-section tension, beside the hole	$\sigma_t = F_{lug}/((W - d_h)\cdot t_{lug})$
Bearing of the pin on the lug	$\sigma_{br} = F_{lug}/(d_p\cdot t_{lug})$
Tear-out shear beyond the hole, on two planes	$\tau_{to} = F_{lug}/(2\cdot a\cdot t_{lug})$
Pin shear, in double shear	$\tau_p = F_{lug}/(2\cdot\pi\cdot d_p^2/4)$
Pin bending moment, from the lever arm across the lug, the gap and the cheek plate	$M = (F_{lug}/2)\cdot(t_{lug}/4 + g + t_s/2)$
Pin bending stress	$\sigma_b = 32\cdot M/(\pi\cdot d_p^3)$
Utilisation of each mode, its stress over the allowable entered	$u = \sigma/\sigma_{allow}$

Basis: Pinned connections[1](Ch. 3); Shear, bearing and bending stresses[2](Ch. 3, 11th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
F_{lug}	Load on the lug (along the lug)	50	kN	spec: LOAD-SCHEDULE-01	0.01 … 100000 kN

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
t_{lug}	Lug thickness	20	mm	drawing: DRG-LL-101 rev A	1 … 500 mm
W	Lug width at the hole	100	mm	drawing: DRG-LL-101 rev A	5 … 2000 mm
d_h	Hole diameter	32	mm	drawing: DRG-LL-101 rev A	2 … 1000 mm
a	Edge distance, hole edge to lug end (load direction)	40	mm	drawing: DRG-LL-101 rev A	1 … 2000 mm
d_p	Pin diameter	30	mm	drawing: DRG-LL-101 rev A	2 … 1000 mm
t_s	Cheek plate thickness (each)	12	mm	drawing: DRG-LL-101 rev A	1 … 500 mm
g	Clearance, lug face to cheek plate (each side)	2	mm	drawing: DRG-LL-101 rev A	0 … 100 mm
f_t	Allowable tensile stress, lug	150	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa
f_{br}	Allowable bearing stress, lug	200	MPa	spec: SPEC-MECH-03 design criteria	1 … 3000 MPa
f_v	Allowable shear stress, lug	90	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa
$f_{b,pin}$	Allowable bending stress, pin	250	MPa	spec: SPEC-MECH-03 design criteria	1 … 3000 MPa
$f_{v,pin}$	Allowable shear stress, pin	150	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa

3. Assumptions

A1. The load acts along the lug's axis through the hole centre. (No out-of-plane or side load.)
A2. The pin bears uniformly over the lug thickness and over each cheek plate. (Pin-bending idealisation for a lug between two cheek plates.)
A3. Each cheek plate reacts half the load. (Pin-bending idealisation for a lug between two cheek plates.)
A4. The pin is a close fit, with d_p at least 0.9 d_h. (Uniform-bearing idealisation.)
A5. This is a design calculation, not a thorough examination. (The sheet checks the lug and pin at the stated loads against the allowables entered. LOLER reg 9 and PUWER duties stay with the owner and the competent person.)

4. Working

[14] $k_{fit} = d_p / d_h$
$= 30\ mm / 32\ mm$
$= 0.9375$
Pin-to-hole ratio

[15] $A_{net} = (W - d_h) * t_{lug}$
$= (100\ mm - 32\ mm) \times 20\ mm$
$= 1360\ mm^2$
Net section through the hole
[16] $\sigma_t = F_{lug} / A_{net}$ [1](Ch. 3)
$= 50\ kN / 1360\ mm^2$
$= 36.765\ MPa$
Net-section tension
[17] $A_{br} = d_p * t_{lug}$
$= 30\ mm \times 20\ mm$
$= 600\ mm^2$
Bearing area
[18] $\sigma_{br} = F_{lug} / A_{br}$
$= 50\ kN / 600\ mm^2$
$= 83.333\ MPa$
Bearing of the pin on the hole
[19] $A_{to} = 2 * a * t_{lug}$
$= 2 \times 40\ mm \times 20\ mm$
$= 1600\ mm^2$
Tear-out area (two planes beyond the hole)
[20] $\tau_{to} = F_{lug} / A_{to}$
$= 50\ kN / 1600\ mm^2$
$= 31.25\ MPa$
Tear-out shear
[21] $A_p = 2 * pi * d_p^2 / 4$
$= 2 \times \pi \times (30\ mm)^2 / 4$
$= 1413.7\ mm^2$
Pin area in double shear
[22] $\tau_p = F_{lug} / A_p$
$= 50\ kN / 1413.7\ mm^2$
$= 35.368\ MPa$
Pin shear
[23] $M_p = F_{lug} / 2 * (t_{lug} / 4 + g + t_s / 2)$
$= 50\ kN / 2 \times (20\ mm / 4 + 2\ mm + 12\ mm / 2)$
$= 0.325\ kN·m$
Pin bending moment (uniform bearing on lug and cheek plates)
[24] $Z_p = pi * d_p^3 / 32$
$= \pi \times (30\ mm)^3 / 32$
$= 2650.7\ mm^3$
Pin section modulus
[25] $\sigma_b = M_p / Z_p$
$= 0.325\ kN·m / 2650.7\ mm^3$
$= 122.61\ MPa$
Pin bending stress
[26] $u_t = \sigma_t / f_t$
$= 36.765\ MPa / 150\ MPa$
$= 0.2451$
Net-section utilisation
[27] $u_{br} = \sigma_{br} / f_{br}$
$= 83.333\ MPa / 200\ MPa$
$= 0.41667$
Bearing utilisation

[28] $u_{to} = \tau_{to} / f_v$
= 31.25 MPa / 90 MPa
= 0.34722
Tear-out utilisation
[29] $u_p = \tau_p / f_{v,pin}$
= 35.368 MPa / 150 MPa
= 0.23579
Pin shear utilisation
[30] $u_b = \sigma_b / f_{b,pin}$
= 122.61 MPa / 250 MPa
= 0.49043
Pin bending utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
u_t	0.25	Net-section utilisation
u_{br}	0.42	Bearing utilisation
u_{to}	0.35	Tear-out utilisation
u_p	0.24	Pin shear utilisation
u_b	0.49	Pin bending utilisation (governing)
M_p	0.3250 kN·m	Pin bending moment
σ_t	36.76 MPa	Net-section tension
σ_{br}	83.33 MPa	Bearing stress
τ_{to}	31.25 MPa	Tear-out shear
τ_p	35.37 MPa	Pin shear
σ_b	122.6 MPa	Pin bending stress
$sheet_{notice}$	"This is a design calculation, not a thorough examination. It does not replace the thorough examination by a competent person (LOLER reg 9) or the marking, testing and certification that the regulations require. Signatures are not authenticated by TempestOS: the signed-in names of the author, the checkers and the competent person are as entered into this installation."	Status of this sheet

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Pin bending stress against the allowable pin bending stress	122.6 MPa	250 MPa	0.49	
PASS	Pin bearing against the allowable bearing stress	83.33 MPa	200 MPa	0.42	[1](Ch. 3)
PASS	Tear-out shear against the allowable shear stress	31.25 MPa	90 MPa	0.35	[1](Ch. 3)
PASS	Net-section tension against the allowable tensile stress	36.76 MPa	150 MPa	0.25	[1](Ch. 3)
PASS	Pin double shear against the allowable pin shear stress	35.37 MPa	150 MPa	0.24	

7. Applicability

None.

8. Verification

Template mech.lifting-lug@1.2.0 Content hash 30f3d85b4747aa83 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. American Society of Mechanical Engineers. Design of Below-the-Hook Lifting Devices. ASME BTH-1. ASME.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.